LAB 7: FUZZING FOR INPUT VALIDATION BUGS

Gym Membership Application

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Course: BTech – CSE (AI & ML)
Assigned Repository: L6_L7_21_gym_membership
GitHub Repository:
https://github.com/Preeti1k3/L6_L7_21_gym_membership
Tools Used: Python, Pytest and Hypothesis

1. Aim:

To identify input-validation bugs in the Gym Membership application using fuzz testing with Hypothesis, and to improve the application so that invalid inputs are handled safely.

2. Objective:

The objective is to generate a wide range of inputs for the application functions, identify inputs that cause unexpected crashes, update the application with appropriate input validation, and verify the fixes by running the fuzz tests again.

3. Application Under Test:

The application file gym_membership.py contains three functions:

- parse_plan() separates a membership plan name and duration from a hyphen-separated input.
- calculate_bmi() calculates BMI using weight and height supplied as comma-separated values.
- average_workout() calculates the average of workout durations supplied as space-separated values.

4. Fuzz Testing Setup:

The test file test_processor.py uses Hypothesis and Pytest to test all three application functions. The strategy st.text() generates different text inputs, and st.none() also tests None values. Each test checks whether the function can process the generated input without raising an exception.

Command used:

```
python -m pytest test_processor.py -v
```

5. Initial Test Execution and Analysis:

The original application was first executed with valid sample inputs. The fuzz tests were then run against the original implementation. Hypothesis generated invalid inputs that exposed unhandled exceptions in all three functions.

- parse_plan(): malformed inputs, missing separators, empty duration values, and None caused ValueError or AttributeError.
- calculate_bmi(): malformed comma-separated values, empty fields, non-numeric values, and None caused ValueError or AttributeError.
- average_workout(): empty input caused division by zero, while non-numeric tokens and None caused ValueError or AttributeError.

The initial fuzz test execution resulted in all three tests failing, showing that the application did not safely handle several invalid inputs.

Original Application Output:

```
PS C:\Users\Preeti Kalburgi\Desktop\sem5\SE_L6_L7\L6_L7_21_gym_membership\Lab 7> python gym_membership.py
==== Running parse_plan ====
Enter a plan code (e.g. GOLD-12): GOLD-12
Plan and Months: ('GOLD', 12)

==== Running calculate_bmi ====
Enter weight and height (e.g. 70,1.75): 70,1.75
BMI: 22.9

==== Running average_workout ====
Enter workout minutes separated by spaces (e.g. 45 60 30): 45 60 30
Average Minutes: 45
PS C:\Users\Preeti Kalburgi\Desktop\sem5\SE_L6_L7\L6_L7_21_gym_membership\Lab 7>
```

Initial Fuzz Test Failures:

```
Average Minutes: 45
PS C:\Users\Preeti Kalburgi\Desktop\sem5\SE_L6_L7\L6_L7_21_gym_membership\Lab 7> python -m pytest test_processor.py -v
============================= test session starts =============================
platform win32 -- Python 3.11.9, pytest-9.2.0, pluggy-1.7.0 -- C:\Users\Preeti Kalburgi\AppData\Local\Programs\Python\Python311\python.exe
cachedir: .pytest_cache
hypothesis profile 'default'
rootdir: C:\Users\Preeti Kalburgi\Desktop\sem5\SE_L6_L7\L6_L7_21_gym_membership\Lab 7
plugins: anyio-4.14.2, hypothesis-6.170.15, langsmith-0.14.5
collected 3 items

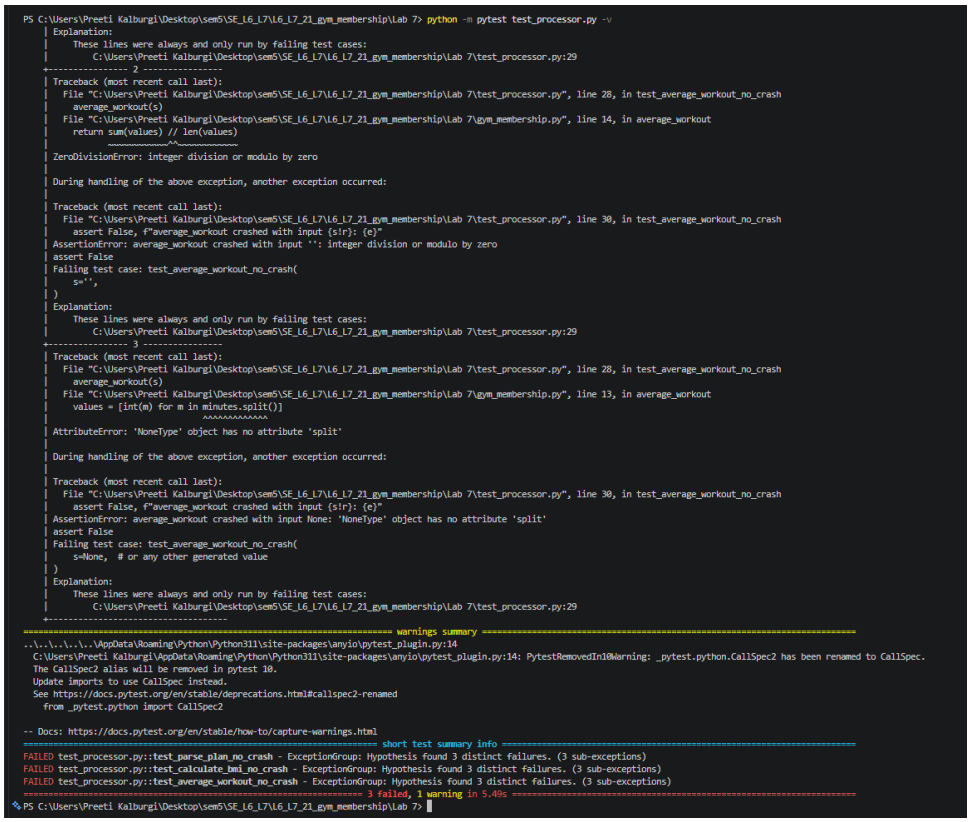
test_processor.py::test_parse_plan_no_crash FAILED                       [ 33%]
test_processor.py::test_calculate_bmi_no_crash FAILED                    [ 66%]
test_processor.py::test_average_workout_no_crash FAILED                  [100%]

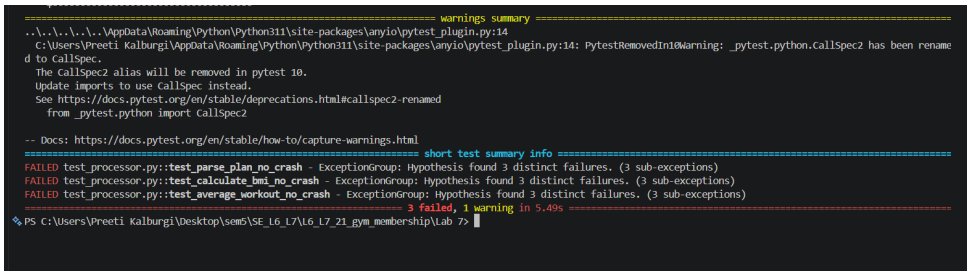
================================== FAILURES ===================================
_____________________________ test_parse_plan_no_crash _____________________________
  + Exception Group Traceback (most recent call last):
  |   File "C:\Users\Preeti Kalburgi\Desktop\sem5\SE_L6_L7\L6_L7_21_gym_membership\Lab 7\test_processor.py", line 8, in test_parse_plan_no_crash
  |     def test_parse_plan_no_crash(s):
  |                ^^^
  | ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "C:\Users\Preeti Kalburgi\Desktop\sem5\SE_L6_L7\L6_L7_21_gym_membership\Lab 7\test_processor.py", line 10, in test_parse_plan_no_crash
    |     parse_plan(s)
    |   File "C:\Users\Preeti Kalburgi\Desktop\sem5\SE_L6_L7\L6_L7_21_gym_membership\Lab 7\gym_membership.py", line 4, in parse_plan
    |     return plan.upper(), int(months)
    |                          ^^^^^^^^^^^
    | ValueError: invalid literal for int() with base 10: ''
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "C:\Users\Preeti Kalburgi\Desktop\sem5\SE_L6_L7\L6_L7_21_gym_membership\Lab 7\test_processor.py", line 12, in test_parse_plan_no_crash
    |     assert False, f"parse_plan crashed with input {s!r}: {e}"
    | AssertionError: parse_plan crashed with input '-': invalid literal for int() with base 10: ''
    | assert False
    | Failing test case: test_parse_plan_no_crash(
    |     s='-',
    | )
    +---------------- 2 ----------------
    | Traceback (most recent call last):
    |   File "C:\Users\Preeti Kalburgi\Desktop\sem5\SE_L6_L7\L6_L7_21_gym_membership\Lab 7\test_processor.py", line 10, in test_parse_plan_no_crash
    |     parse_plan(s)
    |   File "C:\Users\Preeti Kalburgi\Desktop\sem5\SE_L6_L7\L6_L7_21_gym_membership\Lab 7\gym_membership.py", line 3, in parse_plan
    |     plan, months = code.split('-')
```

```
    |     ^^^^^^^^^^
    | ValueError: not enough values to unpack (expected 2, got 1)
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "C:\Users\Preeti Kalburgi\Desktop\sem5\SE_L6_L7\L6_L7_21_gym_membership\Lab 7\test_processor.py", line 12, in test_parse_plan_no_crash
    |     assert False, f"parse_plan crashed with input {s!r}: {e}"
    | AssertionError: parse_plan crashed with input '': not enough values to unpack (expected 2, got 1)
    | assert False
    | Failing test case: test_parse_plan_no_crash(
    |     s='',  # or any other generated value
    | )
    +---------------- 3 ----------------
    | Traceback (most recent call last):
    |   File "C:\Users\Preeti Kalburgi\Desktop\sem5\SE_L6_L7\L6_L7_21_gym_membership\Lab 7\test_processor.py", line 10, in test_parse_plan_no_crash
    |     parse_plan(s)
    |   File "C:\Users\Preeti Kalburgi\Desktop\sem5\SE_L6_L7\L6_L7_21_gym_membership\Lab 7\gym_membership.py", line 3, in parse_plan
    |     plan, months = code.split('-')
    |                    ^^^^^^^^^
    | AttributeError: 'NoneType' object has no attribute 'split'
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "C:\Users\Preeti Kalburgi\Desktop\sem5\SE_L6_L7\L6_L7_21_gym_membership\Lab 7\test_processor.py", line 12, in test_parse_plan_no_crash
    |     assert False, f"parse_plan crashed with input {s!r}: {e}"
    | AssertionError: parse_plan crashed with input None: 'NoneType' object has no attribute 'split'
    | assert False
    | Failing test case: test_parse_plan_no_crash(
    |     s=None,  # or any other generated value
    | )
    +------------------------------------
_____ test_calculate_bmi_no_crash _____
  + Exception Group Traceback (most recent call last):
  |   File "C:\Users\Preeti Kalburgi\Desktop\sem5\SE_L6_L7\L6_L7_21_gym_membership\Lab 7\test_processor.py", line 17, in test_calculate_bmi_no_crash
  |     def test_calculate_bmi_no_crash(s):
  |                    ^^^
  | ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "C:\Users\Preeti Kalburgi\Desktop\sem5\SE_L6_L7\L6_L7_21_gym_membership\Lab 7\test_processor.py", line 19, in test_calculate_bmi_no_crash
    |     calculate_bmi(s)
    |   File "C:\Users\Preeti Kalburgi\Desktop\sem5\SE_L6_L7\L6_L7_21_gym_membership\Lab 7\gym_membership.py", line 9, in calculate_bmi
    |     return round(float(weight) / (float(height) ** 2), 1)
    |                  ^^^^^^^^^^^^
```

```
PS C:\Users\Preeti Kalburgi\Desktop\sem5\SE_L6_L7\L6_L7_21_gym_membership\Lab 7> python -m pytest test_processor.py -v
    |   File "C:\Users\Preeti Kalburgi\Desktop\sem5\SE_L6_L7\L6_L7_21_gym_membership\Lab 7\test_processor.py", line 17, in test_calculate_bmi_no_crash
    |     def test_calculate_bmi_no_crash(s):
    |                    ^^^
    | ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
    +-+---------------- 1 ----------------
      | Traceback (most recent call last):
      |   File "C:\Users\Preeti Kalburgi\Desktop\sem5\SE_L6_L7\L6_L7_21_gym_membership\Lab 7\test_processor.py", line 19, in test_calculate_bmi_no_crash
      |     calculate_bmi(s)
      |   File "C:\Users\Preeti Kalburgi\Desktop\sem5\SE_L6_L7\L6_L7_21_gym_membership\Lab 7\gym_membership.py", line 9, in calculate_bmi
      |     return round(float(weight) / (float(height) ** 2), 1)
      |                  ^^^^^^^^^^^^
      | ValueError: could not convert string to float: ''
      |
      | During handling of the above exception, another exception occurred:
      |
      | Traceback (most recent call last):
      |   File "C:\Users\Preeti Kalburgi\Desktop\sem5\SE_L6_L7\L6_L7_21_gym_membership\Lab 7\test_processor.py", line 21, in test_calculate_bmi_no_crash
      |     assert False, f"calculate_bmi crashed with input {s!r}: {e}"
      | AssertionError: calculate_bmi crashed with input ',': could not convert string to float: ''
      | assert False
      | Failing test case: test_calculate_bmi_no_crash(
      |     s=',',
      | )
      +---------------- 2 ----------------
      | Traceback (most recent call last):
      |   File "C:\Users\Preeti Kalburgi\Desktop\sem5\SE_L6_L7\L6_L7_21_gym_membership\Lab 7\test_processor.py", line 19, in test_calculate_bmi_no_crash
      |     calculate_bmi(s)
      |   File "C:\Users\Preeti Kalburgi\Desktop\sem5\SE_L6_L7\L6_L7_21_gym_membership\Lab 7\gym_membership.py", line 8, in calculate_bmi
      |     weight, height = data.split(',')
      |     ^^^^^^^^^^^^^^
      | ValueError: not enough values to unpack (expected 2, got 1)
      |
      | During handling of the above exception, another exception occurred:
      |
      | Traceback (most recent call last):
      |   File "C:\Users\Preeti Kalburgi\Desktop\sem5\SE_L6_L7\L6_L7_21_gym_membership\Lab 7\test_processor.py", line 21, in test_calculate_bmi_no_crash
      |     assert False, f"calculate_bmi crashed with input {s!r}: {e}"
      | AssertionError: calculate_bmi crashed with input '': not enough values to unpack (expected 2, got 1)
      | assert False
      | Failing test case: test_calculate_bmi_no_crash(
      |     s='',  # or any other generated value
      | )
      +---------------- 3 ----------------
      | Traceback (most recent call last):
      |   File "C:\Users\Preeti Kalburgi\Desktop\sem5\SE_L6_L7\L6_L7_21_gym_membership\Lab 7\test_processor.py", line 19, in test_calculate_bmi_no_crash
      |     calculate_bmi(s)
```

```
PS C:\Users\Preeti Kalburgi\Desktop\sem5\SE_L6_L7\L6_L7_21_gym_membership\Lab 7> python -m pytest test_processor.py -v
    | )
    +---------------- 3 ----------------
    | Traceback (most recent call last):
    |   File "C:\Users\Preeti Kalburgi\Desktop\sem5\SE_L6_L7\L6_L7_21_gym_membership\Lab 7\test_processor.py", line 19, in test_calculate_bmi_no_crash
    |     calculate_bmi(s)
    |   File "C:\Users\Preeti Kalburgi\Desktop\sem5\SE_L6_L7\L6_L7_21_gym_membership\Lab 7\gym_membership.py", line 8, in calculate_bmi
    |     weight, height = data.split(',')
    |                      ^^^^^^^^^
    | AttributeError: 'NoneType' object has no attribute 'split'
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "C:\Users\Preeti Kalburgi\Desktop\sem5\SE_L6_L7\L6_L7_21_gym_membership\Lab 7\test_processor.py", line 21, in test_calculate_bmi_no_crash
    |     assert False, f"calculate_bmi crashed with input {s!r}: {e}"
    | AssertionError: calculate_bmi crashed with input None: 'NoneType' object has no attribute 'split'
    | assert False
    | Failing test case: test_calculate_bmi_no_crash(
    |     s=None,  # or any other generated value
    | )
    +------------------------------------
_____ test_average_workout_no_crash _____
  + Exception Group Traceback (most recent call last):
  |   File "C:\Users\Preeti Kalburgi\Desktop\sem5\SE_L6_L7\L6_L7_21_gym_membership\Lab 7\test_processor.py", line 26, in test_average_workout_no_crash
  |     def test_average_workout_no_crash(s):
  |                    ^^^
  | ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "C:\Users\Preeti Kalburgi\Desktop\sem5\SE_L6_L7\L6_L7_21_gym_membership\Lab 7\test_processor.py", line 28, in test_average_workout_no_crash
    |     average_workout(s)
    |   File "C:\Users\Preeti Kalburgi\Desktop\sem5\SE_L6_L7\L6_L7_21_gym_membership\Lab 7\gym_membership.py", line 13, in average_workout
    |     values = [int(m) for m in minutes.split()]
    |               ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
    |   File "C:\Users\Preeti Kalburgi\Desktop\sem5\SE_L6_L7\L6_L7_21_gym_membership\Lab 7\gym_membership.py", line 13, in <listcomp>
    |     values = [int(m) for m in minutes.split()]
    |               ^^^^^^
    | ValueError: invalid literal for int() with base 10: ':'
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "C:\Users\Preeti Kalburgi\Desktop\sem5\SE_L6_L7\L6_L7_21_gym_membership\Lab 7\test_processor.py", line 30, in test_average_workout_no_crash
    |     assert False, f"average_workout crashed with input {s!r}: {e}"
    | AssertionError: average_workout crashed with input ':': invalid literal for int() with base 10: ':'
    | assert False
```





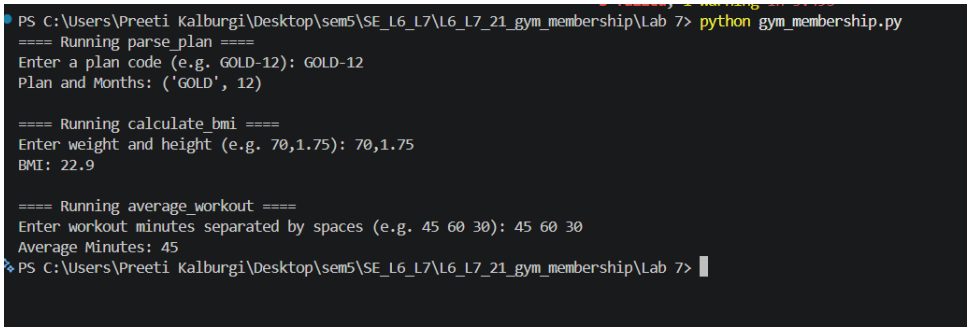
6. Application Fixes:

The application script gym_membership.py was updated to validate inputs before performing conversions and calculations. The functions now check the input type, expected format, and validity of the values.

- parse_plan(): checks that the input is a string with exactly one hyphen, a non-empty plan name, and a positive numeric duration.
- calculate_bmi(): checks the comma-separated input, safely converts weight and height, and verifies that both values are finite and positive before calculating BMI.
- average_workout(): checks that the input is a string containing at least one value and that all workout durations are valid non-negative integers before calculating the average.

For invalid inputs, the fixed functions return None instead of raising an unhandled exception. Valid inputs continue to produce the expected results.

Fixed Application Output:



7. Final Test Execution and Results:

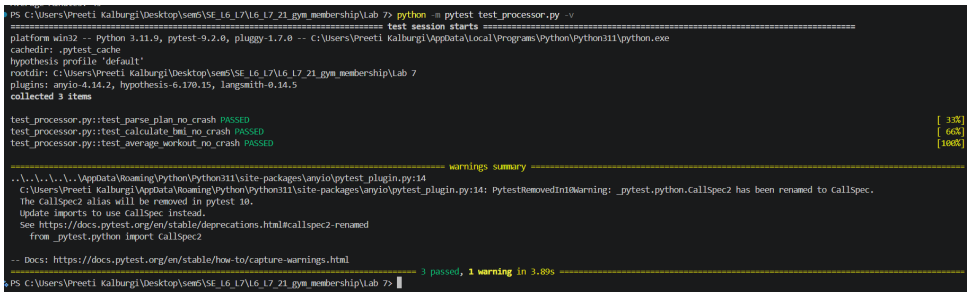
After modifying gym_membership.py, the same Hypothesis fuzz tests in test_processor.py were run again. Reusing the same tests verifies whether the changes resolved the issues identified during the initial execution.

Command used:

python -m pytest test_processor.py -v

All three fuzz tests passed after the input-validation fixes were applied. This confirms that the functions handled the generated test inputs without the previously observed crashes.

Final Passing Test Output:



8. GitHub:

GitHub Repository:
https://github.com/Preeti1k3/L6_L7_21_gym_membership

9. Conclusion:

Fuzz testing was performed on the Gym Membership application using Hypothesis and Pytest. The generated inputs exposed input-validation problems in parse_plan(), calculate_bmi(), and average_workout(). The application was updated to validate input types, formats, and values before processing them. The same fuzz tests were executed again after the fixes, and all three tests passed. This demonstrates that the identified input-handling issues were addressed.